

VELMÈRE BASIC AUDIT REPORT

SafeMoon Protocol Core (0x8076c74c5e3f5852037f31ff0093eeb8c8add8d3)

VELMERE SECURITY AUDIT REPORT: SAFEMOON PROTOCOL CORE

Network: BNB Smart Chain (BSC) (Chain ID: 56)

Report ID: rep_safemoon_basic_046 | Tier: BASIC | Application Surface: CANONICAL

Created At: 2026-09-10T12:36:16.102Z

VERDICT SUMMARY:

CRITICAL RISK (Risk: 88/100 | Quality: 58/100)

Release Decision: BLOCKED | Verification Status: PARTIALLY_VERIFIED

STOP-SELL DIRECTIVE: CRITICAL_RISK_THRESHOLD: Score exceeds maximum institutional tolerance

Confidence Score:	99/100	Evidence Coverage:	97%
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EVM State Provenance: Block #31500000 | Bytecode SHA-256: sha256:88771122a... | DETERMINISTIC_REPRODUCIBLE

VELMERE CRYPTOGRAPHIC AUDIT SEAL - SHA-256 MERKLE ROOT

[VERIFIED - IMMUTABLE]

Root: sha256:30f04b349e4dcfe2...94ec8cd3d4b1



Standard: Merkle Non-Repudiation Seal | RFC 3161 Pinned | Verification: /api/audit/report-pdf

Evidence Coverage Matrix: Bytecode 91% | CFG 86% | Functions 94% | Detectors 100% | Formal Verification 76%

CRITICAL RISK: Legacy SafeMoon v1 contract (0x8076...8add8d3, migrated to v2 0x4298...fcb5). Exhibits centralized owner privilege vectors including arbitrary fee adjustments up to 100% without hard caps. Historical incident: On March 28, 2023, public/unrestricted burn functionality was exploited to arbitrarily burn tokens held by the liquidity pair, artificially inflating the price and draining ~\$8.9M WBNB in the same transaction.

AUDIT OVERVIEW & CONTRACT CONTEXT

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of execution scope and target objectives

Project / Contract: SafeMoon Protocol Core

Contract Address: 0x8076c74c5e3f5852037f31ff0093eeb8c8add8d3

Network / Chain ID: BNB Smart Chain (BSC) (ID: 56)

Token / Symbol: SAFEMOON

Proxy Pattern: Custom Proxy with Unchecked Owner Migration

Compiler Version: solc 0.6.12

SOURCE CODE VERIFICATION & BYTECODE DECOMPIlation

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

State hash match against blockchain node and dispatcher analysis

EVM machine bytecode was disassembled into a control-flow graph (CFG). Standard ERC selectors and privileged opcodes were formally analyzed.

BASELINE SECURITY FINDINGS

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified vulnerabilities in baseline scan

Findings & Vulnerability Analysis:

CRITICAL

VLM-SAFE-01 [SWC-105 / CWE-284]: Public / Unrestricted Burn LP Reserve Desynchronization (\$8.9M Exploit)

Category: Liquidity Pool Manipulation / Arbitrary Burn Vector | State: recommended

Evidence: burn() allows targeting arbitrary holders including Uniswap/PancakeSwap pair contracts

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Recommendation: Quarantine legacy v1 contract; enforce pair address exemption from token destruction and use migrated v2 architecture with immutable liquidity timelocks.

HIGH

VLM-SAFE-02 [SWC-105 / CWE-284]: Centralized Privileged Parameter Risk (Uncapped Fee Setter)

Category: Centralized Governance & Fee Parameter Risk | State: recommended

Evidence: function setTaxFeePercent(uint256 taxFee) external onlyOwner { _taxFee = taxFee; }

Recommendation: Revoke or lock fee setter to maximum 3% hard ceiling.

PRIVILEGED ROLES, ADMIN KEYS & GOVERNANCE

PRO

[LOCKED]

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- Verification of timelock delay parameters and governance controls
- Detection of privileged mint, pause, blacklist, and drainage capabilities
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

LIQUIDITY, HOLDER CONCENTRATION & LOCK EVIDENCE (PRO)

PRO

[LOCKED]

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- Token distribution curve and top holder concentration analysis
- DEX liquidity pool pairing and lock verification across registered vaults
- Automated honeypot simulation, transfer fee checks, and sell restriction audit
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO)

PRO

[LOCKED]

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- Cross-function and cross-contract reentrancy guard modeling
- Price oracle staleness, manipulation resilience, and market feed verification
- Economic shock modeling and flash-loan manipulation resistance simulation
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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STORAGE LAYOUT COLLISION & MULTI-COMPILER BYTECODE DIFF (ADVANCED)

ADVANCED

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Storage slot allocation collision detection across implementation upgrades
- Upgrade proxy layout gap verification and state preservation checks
- Bytecode reproduction against official compiler metadata and build pipelines
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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FORMAL SMT INVARIANT PROOFS & REGRESSION ANALYSIS (ADVANCED)

ADVANCED

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

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Summary of Analytical Depth in this section:

- SMT proofs for solvency, conservation, and reentrancy impossibility invariants
- Bytecode regression analysis and logic change tracking
- Formal SMT Verification (Z3): 76% invariants verified (QF_LIA UNSAT)
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
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HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED)

ADVANCED

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Analyst sign-off requires verified reviewer evidence
- Cryptographically signed attestation digest
- Manual review boundary explicit: no certified safe claims
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
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